

Sex and the Moon

Sexual-activity logging and lunar phase

Boomer Rawlings
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Overview

Do people have sex more often at particular phases of the Moon? This study examines a narrower, measurable question: whether sexual-activity entries in the Clue tracking app follow a lunar pattern after selected calendar differences are taken into account.

The selected model detects a small association: the fitted highest-to-lowest difference across the lunar cycle is **1.58%**, and the joint test gives **p = .039**. That is below the nominal .05 threshold. However, the result changes meaningfully under reasonable alternatives for reporting behavior, statistical uncertainty and geographic coverage.

The specific **Full Moon versus New Moon comparison remains inconclusive**. Its estimated difference is +1.02%, with a 95% confidence interval from 0.47% lower to 2.54% higher. These data do not establish a dependable best night, a general-population pattern or a causal effect of the Moon.

What was observed	What was estimated
730 dates · July 2017-June 2019 Five regions in four countries One app dataset	1.58% fitted peak-to-trough difference Joint lunar test: p = .039 Full versus New Moon: inconclusive

Read the measure correctly

The outcome concerns **optional app logging**. The 1,812,110 qualifying feature logs are neither verified sexual acts nor 1,812,110 independent people. Complete records of days with and without sex are unavailable. The source release accompanies a seasonal-fertility preprint; this lunar analysis is exploratory.^{1,2}

Interactive analysis and downloads: boomerrawlings.com

Reading guide: data and measurement, p. 2; principal results, p. 3; calendar and regional context, p. 4; sensitivity checks, p. 5; statistical methods, pp. 6-7; references, p. 8.

What the data can measure

All five compatible older-band regional series were included, covering 1 July 2017 through 30 June 2019. Each region contributes the same 730 dates. The analysis therefore contains 3,650 region-days and gives each region equal total weight. These are five geographic series from **one source**, not five independent studies.²

Region	Dates	Qualifying sex-feature logs
Brazil: Central-West	730	553,078
France	730	376,716
United Kingdom	730	515,258
United States: California	730	157,227
United States: Northeast	730	209,831
Total	730 distinct	1,812,110

Numerator and denominator

The numerator adds protected-sex, unprotected-sex and withdrawal feature entries. Multiple categories can apply to the same person-day, while repeated acts of a single category are not separately counted. The main denominator counts eligible users with *any* feature attributed to that date. It is not everyone opening the app and does not establish that nonreporters had no sex.^{2,3}

The alternative denominator counts algorithm-defined active users: those with at least two tracking days in a right-aligned 42-day window. It describes a different reporting measure. Neither denominator turns optional logs into complete individual sexual histories.³

Age, dates and missing coverage

The source's older-age label derives from midpoints of five-year birth bins. Selected bins are adult-only conditional on accurate supplied birth information; individual ages are not verified, and the sample is not an exact age-24-and-older population. Brazil Northeast's older stratum is absent. It was not treated as zero or reconstructed from overlapping totals.²

Entries may be added retrospectively. A separate study of the same app documents backfilling, but it does not establish the calendar-date, time-zone or date-shifting contract of this export. The release's calendar labels are usable for an exploratory linkage; actual event timing and upstream date semantics remain unverified.⁴

Why other studies were not averaged in

The source search did not yield another acquired, compatible calendar-qualified diary dataset. Desire scores, monthly recall, birth records, relative study-day numbers and overlapping app totals cannot be combined as if they measured the same daily outcome. The resulting analysis is one-source exploration, not a meta-analysis.

A small fitted pattern, substantial uncertainty

The two lunar terms jointly test a sinusoidal pattern across the entire cycle. They do not ask only whether Full Moon differs from New Moon. A joint test can cross .05 while that particular two-phase contrast does not. A p-value describes evidence against the model's no-pattern hypothesis; it is not the probability that the Moon causes a change.

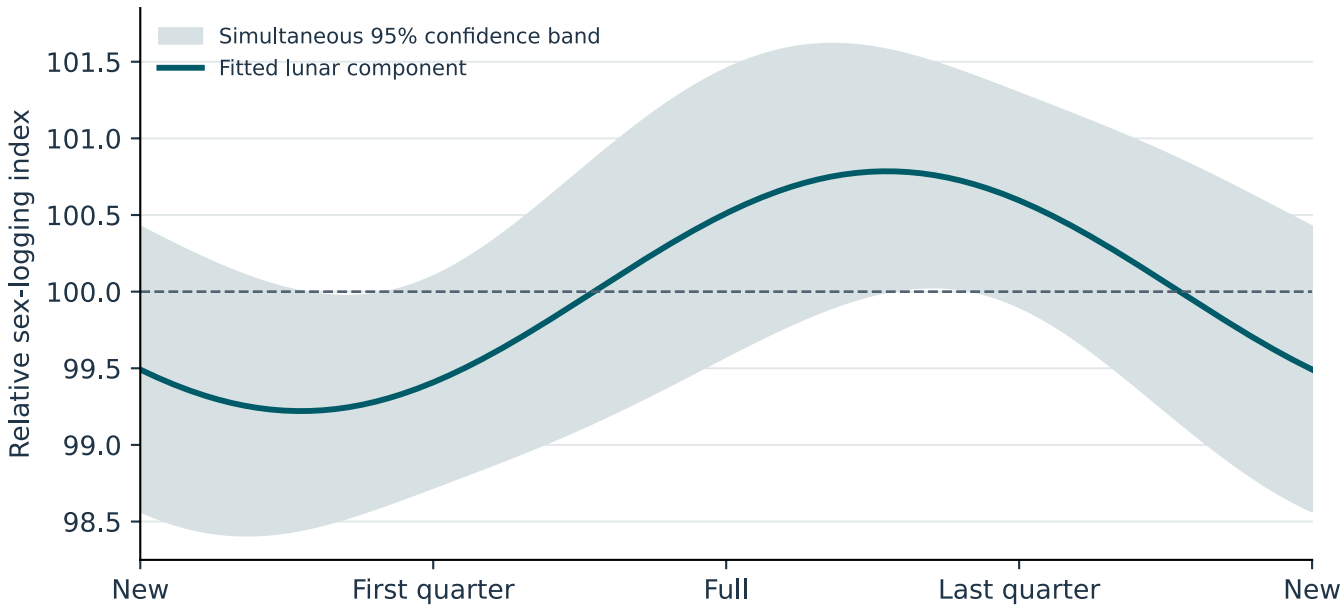


Figure 1. Calendar-adjusted lunar component. An index of 100 is the phase-neutral geometric reference, not 100% of people having sex. Shading is a simultaneous 95% band from the two-coefficient confidence ellipse. The narrow vertical scale makes this small fitted difference visible.

Quantity	Saved estimate	Interpretation
Whole-cycle joint test	p = .039	Nominal association in the selected exploratory model.
Dependent bootstrap	p = .026	Approximate calibration; 1,999 null replicates.
Peak-to-trough difference	1.58% 95% interval: 0.06% to 3.26%	Relative reporting-ratio difference; selected amplitude can be upward-biased.
Full versus New Moon	+1.02% 95% interval: -0.47% to +2.54%	The interval includes no difference.
Extra residual variation explained	0.39%	After the specified calendar terms; a small additional contribution.

The fitted peak is near 229°, in the waning-gibbous part of the cycle. Its joint confidence region allows peak directions from roughly 166° to 320°. That wide range, together with the sensitivity results, does not identify a precise most-likely night.

The peak-to-trough interval is a projection of the joint confidence ellipse. The Full-versus-New interval concerns its specified contrast. All figures are changes in the reporting ratio, not percentage-point changes in the proportion of people having sex.

Calendar and geographic context

Ordinary weekly reporting differences are much larger descriptively than the lunar-phase differences. The panels below share a vertical scale. They show unadjusted geometric group averages, without inferential confidence intervals; the formal test uses continuous phase and calendar adjustment.

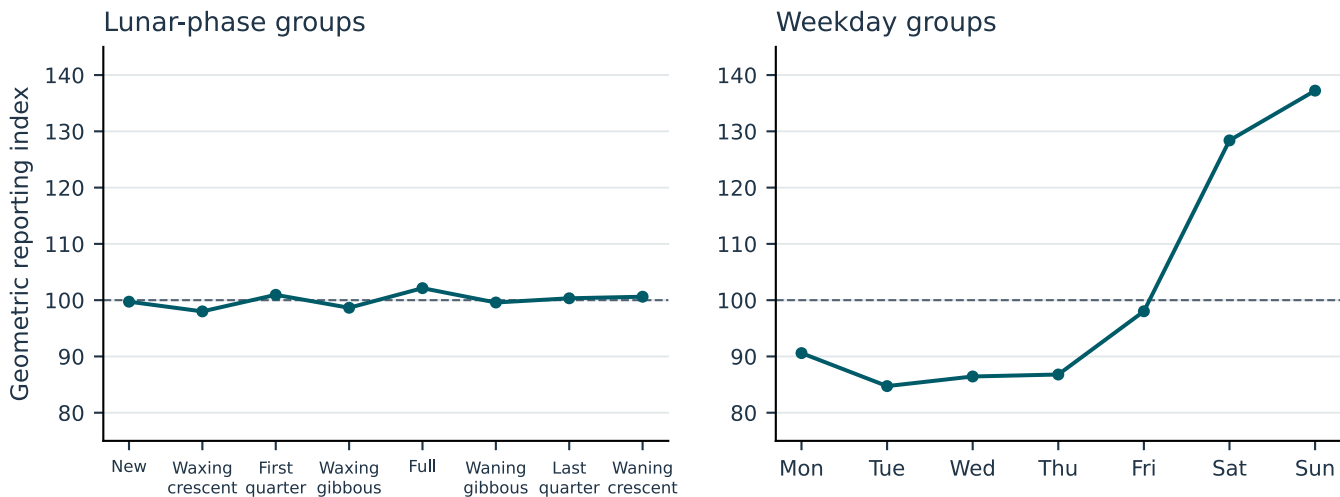


Figure 2. Descriptive reporting indices, with each region centered at 100 before pooling. Unequal phase-bin sizes and overlap with calendar effects matter. These averages describe logging, not individual sexual probabilities.

Regional estimates

Region	Peak-to-trough	Joint p	Holm-adjusted p
Brazil Central-West	1.43%	.505	1.000
France	0.86%	.756	1.000
United Kingdom	2.31%	.120	.359
United States: California	2.71%	.013	.054
United States: Northeast	3.42%	.0016	.0079

The five regional tests use the prespecified Holm correction for multiple comparisons. Their selected amplitudes do not share a common peak direction. The Northeast result merits follow-up, but making that subgroup the headline would overstate the pooled evidence.

These regions share the same astronomical calendar and app measurement process. They are geographic descriptions, not independent replications. A region's joint p-value is also a different question from its Full-versus-New contrast.

How stable is the result?

Checks in the following table were fixed before the first lunar-phase result. The main model was not replaced after reviewing their p-values. The full 21-model table is available with the interactive analysis.

Check	Joint lunar p	What it shows
Main: 35-day dependence allowance	.039	Selected exploratory specification.
14-day / 60-day allowances	.146 / .013	Same fitted curve; materially different uncertainty.
Four annual harmonics instead of two	.039	Little change with richer seasonality terms.
Representative time shifted by ±12 hours	.038-.040	Stable to this convention; actual timestamps remain unverified.
Rolling-active-user denominator	.060	Different measure; Full-versus-New changes to -0.25%.
Exclude California / Northeast	.195 / .198	Pooled evidence weakens.
Exclude Brazil / France / United Kingdom	.0013 / .021 / .054	Geographic composition affects the pooled result.
First / second study year	.127 / .425	Fitted amplitudes: 2.31% / 0.76%.
Calendar-unadjusted model	.293	Similar amplitude; more unexplained variation.

These checks support a **small, model-dependent exploratory association**. Separate year fits are less precise: failing to cross .05 in either year does not itself prove that their effects differ. Leave-one-region-out p-values are sensitivity descriptions, not tests of regional heterogeneity.

Reporting behavior also matters. The share of rolling active users with any tracked feature has a lunar association in the same model ($p = .017$). The fitted coefficients obey the exact identity linking this tracking fraction and the two sex-logging denominators. This is a measurement diagnostic, not replication or proof that logging explains the entire pattern.

Separate post hoc calendar check

After the main results were known, the source audit identified an omitted Brazil-only June 12 dating-holiday indicator. The original source models that celebration; a municipal source confirms its date. ^{2,10}

Adding that single term gives $p = .032$ and a 1.62% fitted peak-to-trough difference. Full versus New Moon is +1.06% (95% interval: -0.47% to +2.61%). This later check does not replace the original $p = .039$ result or the 21 planned outputs, and it does not make the calendar adjustment exhaustive.

Model, phase and calendar adjustment

The model is an unweighted ordinary least-squares regression of the logarithm of the reporting ratio. Balanced region-days give the five regions equal weight. All selected counts are positive; no artificial count was added.

$$\log\left(\frac{n_{rt}}{A_{rt}}\right) = \alpha_r + \mathbf{c}_{rt}^T \boldsymbol{\gamma}_r + a \cos(\theta_{rt}) + b \sin(\theta_{rt}) + \varepsilon_{rt}$$

Here, r is region and t is date; n counts qualifying sex-feature logs, A counts users with any attributed feature, and θ is astronomical phase. The region intercept α and calendar terms $\mathbf{c}$ have region-specific coefficients. The common lunar coefficients are a and b .

Calendar terms

Each region has six weekday indicators, a centered linear trend and two annual sine/cosine harmonics. Fixed event windows cover Christmas (December 24-26), New Year (December 31-January 1), Valentine's Day (February 14), Western Easter (Friday-Monday), Brazil Carnival (Monday-Tuesday), relevant national dates and United States Thanksgiving (Thursday-Friday). National dates are July 4, July 14 and September 7 in the applicable regions.

These are selected controls, not a complete holiday inventory. Easter and Carnival are linked to a lunar-derived religious calendar; adjusting for them does not isolate a biological mechanism. Individual menstrual cycles, relationship status, weather, artificial light and partner availability cannot be controlled from these aggregate fields.

Astronomical exposure

Phase is the apparent geocentric ecliptic longitude of the Moon minus that of the Sun, calculated with Skyfield and the National Aeronautics and Space Administration / Jet Propulsion Laboratory (NASA/JPL) DE421 ephemeris. New Moon is 0° , first quarter 90° , Full Moon 180° and last quarter 270° . It is not measured local moonlight or illuminated-surface percentage.^{5,6}

Representative local noon uses historical time-zone rules for Paris, London, Los Angeles, New York and São Paulo/Brasília time. Those conventions approximate broad regions; Brazil Central-West spans multiple zones. The ± 12 -hour check assesses that convention, not the accuracy of event dating.

All 50 primary-phase crossings in 2018 agreed within 34 seconds of published, minute-resolution United States Naval Observatory (USNO) times. This checks the astronomical implementation; it does not establish subminute event-time accuracy. The observed window spans about 25 lunar cycles and contains 24 New Moon crossings.⁷

Uncertainty, interpretation and audit

The main joint test evaluates both lunar coefficients together against zero, using a two-degree-of-freedom Wald test at a nominal .05 threshold. Date-summed scores and Bartlett heteroskedasticity/autocorrelation-consistent (HAC) covariance allow all regions to share date shocks. The main bandwidth is 35 days; 14 and 60 days are fixed sensitivity checks.⁸

$$H_0 : a = b = 0, \quad f = \frac{T}{T-1} \frac{N-1}{N-K}$$

The covariance's finite-sample multiplier is f , $T = 730$ dates, $N = 3,650$ region-days and $K = 89$ fitted parameters. This correction does not create additional independent lunar exposures.

Effect sizes

$$\Delta_{\text{peak/trough}} = 100[e^{2\sqrt{a^2 + b^2}} - 1], \quad \Delta_{\text{full/new}} = 100[e^{-2a} - 1]$$

The simultaneous curve band and projected peak-to-trough interval come from the joint two-coefficient confidence ellipse. The Full-versus-New interval uses its specified linear contrast before exponentiation. A fitted maximum is selected from the curve, so amplitude tends to be upward-biased when the underlying signal is weak.

Bootstrap calibration

The null-imposed dependent wild bootstrap uses 1,999 replicates, 35-day moving-average Gaussian multipliers and a shared multiplier for each date across all regions. Every replicate refits the full model and covariance. Its p-value is $(1 + 51)/(1 + 1,999) = .026$. This is an approximate check under time-series and nuisance-model assumptions, not an exact randomization test.⁹

No independent-day shuffle or arbitrary phase rotation served as a null test. Neither the covariance nor the bootstrap treats millions of logs as millions of independent lunar observations.

Verification and remaining limits

A separate implementation rebuilt all 21 models, calendars, coefficients, covariance estimates, intervals, regional corrections and bootstrap. The largest difference in a joint p-value was below 5×10^{-13} ; an additional statsmodels calculation agreed. The post hoc holiday fit was checked separately. A separate source audit reconciled measurement, astronomy and source-to-claim use.

The analysis plan was locally saved before lunar outcome inspection, but was **not externally preregistered**. Numerical agreement verifies implementation, not self-report completeness, date semantics, representativeness or causation. Independent, complete dated diaries and additional years are needed before drawing conclusions about general sexual behavior.

Public outputs: [all 21 model results](#); [adjusted curve and band](#); [source register](#). Only derived aggregate outputs accompany this report; no participant records are released.

References and source locators

Numbered citations link to these entries. Return links navigate back to each use in the report. Primary source versions and code locators distinguish measurement evidence from the new calculations.

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4. **Pierson E, Althoff T, Thomas D, Hillard P, Leskovec J. Daily, weekly, seasonal and menstrual cycles in women's mood, behaviour and vital signs.** Nature Human Behaviour. 2021;5:716-725. DOI: 10.1038/s41562-020-01046-9. Methods: 'Clue app interface'. [Source](#). Retrospective, once-per-day feature entry; not this export's date-transformation contract. [return 1](#)
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9. **Shao X. The Dependent Wild Bootstrap.** Journal of the American Statistical Association. 2010;105(489):218-235. DOI: 10.1198/jasa.2009.tm08744. [Source](#). Dependent bootstrap method family and assumptions; application here is approximate. [return 1](#)
10. **Municipality of Mundo Novo, Mato Grosso do Sul. O Dia dos Namorados é comemorado em 12 de junho no Brasil.** Published 12 June 2023. Official municipal explanation of the Brazilian June 12 celebration. [Source](#). Date of the separately disclosed post hoc holiday indicator. [return 1](#)